

International Diversification, Reallocation, and the Labor Share

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September 14, 2022

Motivating Fact

↓ Labor share of income.¹

↑ Foreign holdings of domestic risky assets (e.g., equity).

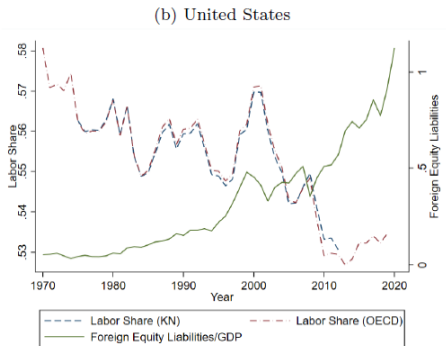


Figure 1: Trends in Labor Share and Equity Share of Foreign Investors

¹Karabarounis and Neiman, 2014

Overview

This paper:

Proposes a link between international financial integration, risk sharing and the labor share.

Theoretical Framework:

Firms insure workers and are exposed to domestic market risk.

Riskier firms $\Rightarrow$ Lower firm labor share

Financial integration reduces domestic market risk.

- Every firm will increase the share of income paid to labor.
- Reallocation of labor toward riskier firms.

The second effect was stronger in the past 40 years.

Empirical analysis:

Document a number of empirical patterns consistent with the model:

- Riskier firms have lower labor shares.
- International diversification $\Rightarrow$ Reallocation towards risky firms
- International diversification $\Rightarrow$ Declines in the aggregate labor share
- Industries with greater heterogeneity have higher sensitivity of their labor share to international diversification.

Theoretical Framework: Risk and the Labor Share

Risk, Input Allocations and the Labor Share

Static problem.

Partial equilibrium economy with exogenous SDF: Λ

Multiple firms, each producing according to:

$$Y_i = A_i K_i^{\alpha_1} L_i^{\alpha_2}, \quad \alpha_1 + \alpha_2 < 1$$

A_i is **stochastic**, and $\text{Cov}(A_i, \Lambda) \leq 0$.

Input choices made prior to the realization of A_i .

Firms maximize the expected discounted cash flow:

$$\max_{K_i, L_i} E[\Lambda(A_i K_i^{\alpha_1} L_i^{\alpha_2} - WL_i - RK_i)]$$

Risk, Input Allocations and the Labor Share

Optimal condition for labor:

$$E[\Lambda(MPL_i)] = E[\Lambda]W$$

Rearrange:

$$\frac{WL_i}{E[Y_i]} = \frac{E[(MPL_i)]L_i}{E[Y_i]} \left(1 + \frac{Cov(\Lambda, MPL_i)}{E[\Lambda]E[MPL_i]} \right)$$

Or

$$\frac{WL_i}{E[Y_i]} = \alpha_2(1 - \kappa_i)$$

where κ_i is the *risk premium*

$$\kappa_i = -\frac{Cov(\Lambda, A_i)}{E[\Lambda]E[A_i]}$$

Risk, Input Allocations and the Labor Share

The expected aggregate labor share is given by:

$$\frac{WL}{E[Y]} = \sum_i \frac{E[Y_i]}{E[Y]} \frac{WL_i}{E[Y_i]}$$

It can be show that the output share of each firm is:

$$\frac{E[Y_i]}{E[Y]} = \frac{E[A_i]^{\frac{1}{1-\alpha_1-\alpha_2}} (1 - \kappa_i)^{\frac{\alpha_1+\alpha_2}{1-\alpha_1-\alpha_2}}}{\sum_h E[A_h]^{\frac{1}{1-\alpha_1-\alpha_2}} (1 - \kappa_h)^{\frac{\alpha_1+\alpha_1}{1-\alpha_1-\alpha_2}}}$$

So, the aggregate labor share is:

$$\frac{WL}{E[Y]} = \frac{\sum_i (E[A_i](1 - \kappa_i))^{\frac{1}{1-\alpha_1-\alpha_2}}}{\sum_h E[A_h]^{\frac{1}{1-\alpha_1-\alpha_2}} (1 - \kappa_h)^{\frac{\alpha_1+\alpha_1}{1-\alpha_1-\alpha_2}}}$$

A Simple Case - Two Firms

Suppose two firms: $\{s, r\}$.

Risky firm has stochastic productivity and $Cov(A_r, \Lambda) < 0$

The risky firm requires risk premium κ_r and so:

$$\frac{WL_s}{E[Y_s]} = \alpha_{s2} \quad \text{and} \quad \frac{WL_r}{E[Y_r]} = \alpha_{r2}(1 - \kappa_r)$$

In this setting, we have:

$$\frac{\partial \frac{WL}{E[Y]}}{\partial \kappa_r} = \underbrace{\sum_i \frac{E[Y_i]}{E[Y]} \frac{\partial \frac{WL_i}{E[Y_i]}}{\partial \kappa_r}}_{\text{within effect}} + \underbrace{\sum_i \frac{\partial \frac{E[Y_i]}{E[Y]}}{\partial \kappa_r} \frac{WL_i}{E[Y_i]}}_{\text{reallocation effect}}$$

A Simple Case - Two Firms

Proposition 1

A decrease in κ_r implies:

1. The within effect increase the labor share:

$$\sum_i \frac{E[Y_i]}{E[Y]} \frac{\partial \frac{WL_i}{E[Y_i]}}{\partial \kappa_r} = -\alpha_2 \frac{E[Y_r]}{E[Y]} < 0$$

2. The reallocation effect reduces the labor share:

$$\sum_i \frac{\partial \frac{E[Y_i]}{E[Y]}}{\partial \kappa_r} \frac{WL_i}{E[Y_i]} = \alpha_2 \frac{\alpha_1 + \alpha_2}{1 - \alpha_1 - \alpha_2} \frac{E[Y_r]}{E[Y]} \frac{E[Y_s]}{E[Y]} \frac{\kappa_r}{1 + \kappa_r} > 0$$

3. There is a threshold $\bar{\kappa}_r$ such that the labor share is increasing in κ_r : $\frac{\partial \frac{WL}{E[Y]}}{\partial \kappa_r} > 0$

A Simple Case - Illustration

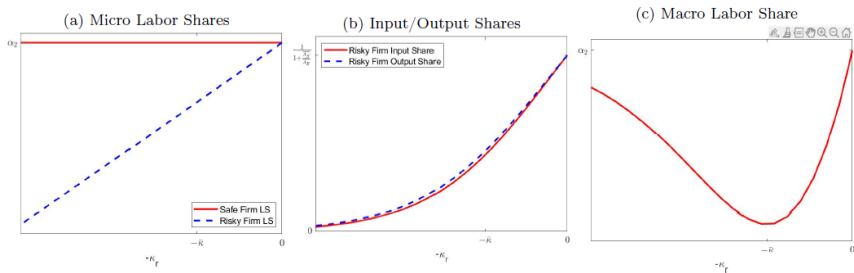


Figure 2: The Price of Risk and the Labor Share

International Diversification, the Risk Premium, and the Labor Share

Full Model

Two period model $t \in \{0, 1\}$. Production only happens in period 1.

Continuum of countries indexed by j , each with two type of firms $i \in \{s, r\}$

Homogeneous tradable consumption goods.

Two type of Households: **Workers** and **Capitalists**

Workers decide how much labor to supply in period 0, work and consume in period 1.

Workers have standard preferences over consumption and leisure.

Budget constraint:

$$C_{wj} = W_j \sum_i L_{ij}$$

Full Model - Capitalists

Capitalists don't work, start with endowment K_0 .

Consume in both periods and have budget constraint.

$$C_{ej0} = K_{0j} - K_{1j} - \sum_i S_{i,j} P_{ij} - \sum_i \sum_{h \neq j} (1 + \tau_h) S_{i,h,j} P_{ih} - P_{rf} S_{rf} + T_{0,j}$$

$$C_{ej} = \sum_i S_{i,j} V_{i,j} + \sum_{h \neq j} \sum_i S_{i,j,h} V_{i,h} + S_{rf} + P_{kj} K_{1j}$$

Where:

C_{ej0}, C_{ej} : Consumption in period 0 and 1.

K_{1j} : Capital saved for period 1.

$S_{i,j}$: Share of a firm type i from country j (home country)

$S_{i,h,j}$: Share of a firm type i from country h .

τ_h : Cost of investing in country h

S_{rf} : Amount of risk free bonds ($S_{rf} = 0$)

Full Model - Capitalists

Capitalists don't work, start with endowment K_0 .

Consume in both periods and have budget constraint.

$$C_{ej0} = K_{0j} - K_{1j} - \sum_i S_{i,j} P_{ij} - \sum_i \sum_{h \neq j} (1 + \tau_h) S_{i,h,j} P_{ih} - P_{rf} S_{rf} + T_{0,j}$$

$$C_{ej} = \sum_i S_{i,j} V_{i,j} + \sum_{h \neq j} \sum_i S_{i,j,h} V_{i,h} + S_{rf} + P_{kj} K_{1j}$$

Where:

$$T_{0,j} = \sum_i S_{i,j} P_{ij} + \sum_i \sum_{h \neq j} (1 + \tau_j) S_{i,j,h}^* P_{ij}$$

$$V_{i,j} = A_{i,j} K_{i,j}^{\alpha_1} L_{i,j}^{\alpha_2} - K_{i,j} P_{k,j} - W_j L_{i,j}$$

Preferences are CRRA over consumption and $E[\Lambda] = \frac{U'_2(C_{ej})}{U'_1(C_{ej0})}$

Full Model - Integration and Diversification

τ_j is the cost investors have to pay to invest in the home country.

When both foreign and home investors hold capital of firm $i = r$, we need:

$$P_{r,j} = E[\Lambda_j V_{r,j}], \quad P_{r,j} = \frac{E[\Lambda_h V_{r,j}]}{(1 + \tau_j)}$$

Assume $E[\Lambda_h] = 1$, $\text{Cov}(\Lambda_h, V_{r,j}) = 0$.

Combine both expressions:

$$\kappa_{r,j} = -\frac{\text{Cov}(\Lambda, A_{r,j})}{E[\Lambda_j]E[A_{r,j}]} = \alpha_2 \frac{\tau_j(1 - \alpha_1 - \alpha_2)}{1 - \tau_j(1 - \alpha_1 - \alpha_2)}$$

Therefore, $\downarrow \tau_j \Rightarrow \downarrow \kappa_{r,j}$ for $\tau_j \in (0, \tau^{aut}]$

Full Model - Diversification and Labor Share

Using $\kappa_{r,j}$ in the equation for the risky firm Labor Share:

$$\frac{L_{rj} W_j}{E[Y_{r,j}]} = \alpha_2 \frac{1}{1 + \tau_j(1 - \alpha_1 - \alpha_2)}$$

And the share of labor allocated to the risky firm:

$$\frac{L_{rj}}{L_j} = \frac{\left(E[A_{rj}] \frac{1}{1 + \tau_j(1 - \alpha_1 - \alpha_2)} \right)^{\frac{1}{1 - \alpha_1 - \alpha_2}}}{E[A_{sj}]^{\frac{1}{(1 - \alpha_1 - \alpha_2)}} + \left(E[A_{rj}] \frac{1}{1 + \tau_j(1 - \alpha_1 - \alpha_2)} \right)^{\frac{1}{1 - \alpha_1 - \alpha_2}}}$$

Full Model - Integration and Labor Share

Proposition 3

A decrease in the cost of foreign investment, τ_j , implies:

1. The equilibrium price of risk decreases.
2. Domestic holdings of the risky firm fall.
3. The within effect increase the labor share: $\sum_i \frac{\partial \frac{E[Y_i]}{E[Y]}}{\partial \tau_r} < 0$
4. The reallocation effect decreases the labor share:

$$\sum_i \frac{E[Y_i]}{E[Y]} \frac{\partial \frac{WL_i}{E[Y_i]}}{\partial \tau_r} > 0$$

5. There is a threshold $\bar{\tau}_j$ such that: $\frac{\partial \frac{WL}{E[Y]}}{\partial \tau_j} > 0$, iff $\tau_j > \bar{\tau}_j$

Empirical Predictions

International Diversification and the Labor Share

Prediction 1: International diversification affects aggregate labor labor share.

Expanding the equation for aggregate labor share:

$$\log \frac{W_j L_j}{Y_j} = \alpha_j + \gamma_\tau \tau_j + \gamma_{tfp} (tfp_j - E[tfp_j])$$

Suggesting the specification:

$$\log \frac{W_{j,t} L_{j,t}}{Y_{j,t}} = \alpha_j + \gamma_{FEQ} FEQ_{j,t} + \gamma_{tfp} (tfp_j - E[tfp_j]) + \varepsilon_{j,t}$$

FEQ: Value of equity held by foreign investors / GDP

Prediction: $\gamma_{FEQ} < 0$

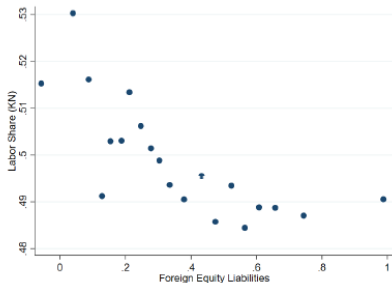
International Diversification and the Labor Share

Data:

Labor Share: Karabarbounis and Neiman (2014), OCDE

FEQ: External Wealth Nations dataset of Lane and Milesis-Ferretti (2018)

(a) Karabarbounis and Neiman (2014)



(b) OECD

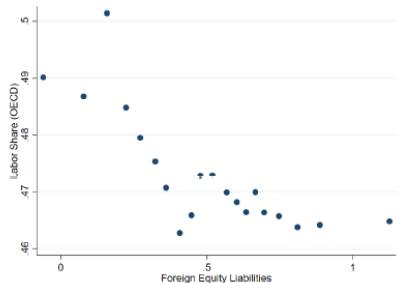


Figure 3: Foreign Equity Liabilities and the Labor Share

International Diversification and the Labor Share

Table 1: International Diversification and the Labor Share

	(1)	(2)	(3)	(4)
Foreign Equity Liabilities	-0.0685** (-2.51)	-0.0491** (-2.62)	-0.0283** (-2.30)	-0.0387*** (-3.88)
TFP shock			-0.237 (-1.17)	-0.397** (-2.18)
Average hours				-0.127 (-0.90)
Relative price of investment				0.102*** (3.38)
Country fixed effects	yes	yes	yes	yes
Year fixed effects	no	yes	yes	yes
R^2	0.853	0.878	0.921	0.937
Observations	661	661	423	402

Between 1975 - 2012: Around 1.1% decline in the labor share.

Risk and Share Paid to Labor

Prediction 2: Risky firms have lower labor share

The equation for individual firm labor share suggests:

$$\log LS_{ij} = \gamma_{\beta} \beta_i + \text{controls} + \varepsilon_{j,t}$$

Theory predicts: $\gamma_{\beta} < 0$

Data:

Betas: Firm level data from Compustat.

Labor Share: Compustat

- Employees divided by sales: $\frac{L}{Y}$
- Compensation divided by sales: LS - (only 9% of the sample)
- Extended labor share: ELS - Infer compensation of firms with no data from the industry average of those who report.

Risk and Share Paid to Labor

Table 2: Firm-Level Labor Shares and Risk Exposures

	(1) $\log\left(\frac{L}{Y}\right)$	(2) $\log(ELS)$	(3) $\log(LS)$	(4) $\log\left(\frac{L}{Y}\right)$	(5) $\log(ELS)$	(6) $\log(LS)$
$\hat{\beta}$	-0.238*** (-12.57)	-0.241*** (-16.24)	-0.105*** (-3.16)	-0.336*** (-16.35)	-0.176*** (-14.62)	-0.166*** (-5.77)
Profitability				-1.019*** (-24.10)	-1.514*** (-33.93)	-1.470*** (-20.92)
Age				-0.0275*** (-5.85)	-0.00781*** (-2.73)	-0.00738 (-1.14)
Size				0.0609*** (14.69)	-0.00607*** (-2.74)	0.0253*** (4.36)
Industry-year F.E.	yes	yes	yes	yes	yes	yes
R^2	0.677	0.405	0.718	0.716	0.510	0.797
Observations	153676	126730	11536	142760	118455	10039

Notes: Table presents regressions of (log) firm labor shares on firm betas and controls. Standard errors are two-way clustered by firm and year. t -statistics in parentheses. Significance levels are denoted by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Diversification and Reallocation

Prediction 3: Int. diversification $\Rightarrow$ Reallocation towards risky/Low labor share firms

These results suggest:

$$\log \frac{Z_{it}}{\bar{Z}_t} = \gamma_{\beta, FEQ}(\beta_i \times FEQ_t) + controls + \varepsilon_{j,t}$$

$$\log \frac{Z_{it}}{\bar{Z}_t} = \gamma_{LS, FEQ}(LS_{it} \times FEQ_t) + controls + \varepsilon_{j,t}$$

Where $Z \in \{L, Y, K\}$ and $\bar{Z}$ denote industry average

Theory predicts: $\gamma_{\beta, FEQ} > 0$ and $\gamma_{LS, FEQ} < 0$

Risk and Share Paid to Labor

Table 3: International Diversification and Reallocation to Risky Firms

	(1) $\log(\frac{Y_i}{Y_{ind}})$	(2) $\log(\frac{L_i}{L_{ind}})$	(3) $\log(\frac{K_i}{K_{ind}})$	(4) $\log(\frac{Y_i}{Y_{ind}})$	(5) $\log(\frac{L_i}{L_{ind}})$	(6) $\log(\frac{K_i}{K_{ind}})$
$\hat{\beta} \times \text{FEQ}$	2.318*** (11.29)	1.804*** (9.11)	2.372*** (9.44)	2.324*** (10.59)	1.829*** (8.82)	2.392*** (9.02)
Profitability				1.479*** (34.38)	0.908*** (25.86)	1.058*** (26.56)
Age				0.278*** (21.46)	0.247*** (18.69)	0.233*** (14.33)
Industry-year F.E.	yes	yes	yes	yes	yes	yes
Firm F.E.	yes	yes	yes	yes	yes	yes
R^2	0.943	0.941	0.938	0.950	0.945	0.942
Observations	151652	150763	152913	146330	145021	147208

Notes: Table presents regressions of firm shares of industry output, labor and capital on the interaction of firm betas and US foreign equity liabilities as a fraction of GDP. Standard errors are two-way clustered by firm and year. t -statistics in parentheses. Significance levels are denoted by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Risk and Share Paid to Labor

Table 4: International Diversification and Reallocation to Low Labor Share Firms

	(1) $\log(\frac{Y_i}{Y_{ind}})$	(2) $\log(\frac{L_i}{L_{ind}})$	(3) $\log(\frac{K_i}{K_{ind}})$	(4) $\log(\frac{Y_i}{Y_{ind}})$	(5) $\log(\frac{L_i}{L_{ind}})$	(6) $\log(\frac{K_i}{K_{ind}})$
$\log(ELS) \times \text{FEQ}$	-0.328*** (-5.65)	-0.256*** (-4.44)	-0.406*** (-6.08)	-0.398*** (-6.97)	-0.311*** (-5.69)	-0.447*** (-6.65)
Profitability				0.984*** (19.19)	0.948*** (18.56)	0.550*** (8.74)
Age				0.237*** (19.37)	0.238*** (20.14)	0.251*** (16.06)
Industry-year F.E.	yes	yes	yes	yes	yes	yes
Firm F.E.	yes	yes	yes	yes	yes	yes
R^2	0.954	0.942	0.947	0.957	0.945	0.949
Observations	119414	118338	118994	116579	115551	116180

Notes: Table presents regressions of firm shares of industry output, labor and capital on the interaction of firm labor share and US foreign equity liabilities as a fraction of GDP. Standard errors are two-way clustered by firm and year. *t*-statistics in parentheses. Significance levels are denoted by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Labor Share and Labor Share Dispersion

Prediction 4: Industries with higher risk exposure dispersion (labor share dispersion) should see larger declines in LS

Higher dispersion $\Rightarrow$ More scope for reallocation

$$\log LS_{s,j,t} = \gamma_u \bar{LS}_{sjt} \times FEQ_{jt} + \gamma_\sigma \text{var}(LS_{sjt}) \times FEQ_{jt} + \text{controls} + \varepsilon_{j,t}$$

Industries with higher mean labor share should have a lower within effect.

prediction: $\gamma_u, \gamma_\sigma < 0$.

Data: LS - Orbis data - industry (s), country(j)

Labor Share and Labor Share Dispersion

Table 5: Diversification, Heterogeneity and Industry-Level Labor Share

	(1)	(2)	(3)
Foreign Equity Liabilities $\times$ mean log(LS)	-0.127* (-1.86)	-0.0869** (-2.74)	-0.0940** (-2.62)
Foreign Equity Liabilities $\times$ stdev log(LS)	-0.0983* (-1.95)	-0.0513** (-2.40)	-0.0592** (-2.26)
Foreign Equity Liabilities $\times$ stdev log(sales)			-0.0099 (-1.22)
Foreign Equity Liabilities $\times$ mean log(sales)			-0.0047 (-1.07)
Fixed effects	no	yes	yes
R^2	0.485	0.791	0.804
Observations	71346	69431	57325

Notes: Table presents regression of industry-level labor shares on the mean and standard deviation of (log) labor shares within that industry interacted with foreign equity liabilities as a fraction of GDP. All specifications include the interacted variables individually as controls. Fixed effects, when included, consist of industry-year, country-year and industry-country effects. Standard errors are two-way clustered by industry-country and year. t -statistics in parentheses. Significance levels are denoted by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Conclusion

Develop a theory about the labor share of income with heterogeneous firms making decisions under uncertainty.

In equilibrium, riskier firms pay a lower share of income to labor.

As financial integration reduces the risk premium.

- All firms increase the Labor Share
- Labor migrates from High LS firms to Low LS firms

For lower levels of integration, the second effect dominates.

They argue this has been the case in the past 40 years.

- Document a number of empirical patterns to support this view